



SPACECAM PROJECT

Team Spring

Dynamics, Mission Control, Survivability, and Electronics

Mission Success



Dynamics Predictor Spreadsheet

	SI		AES	
Input parameters	Value	Unit	Value	Unit
Launch altitude above sea level	299.9232	m	984	ft
Load mass	1.216	kg	2.68128	lbm
Balloon size (select one)	600	g	1.322772	lbm
Inflated balloon diameter	1.524	m	5	ft
Balloon fill volume	1.85333328	m³	65.4498469	ft³
Parachute diameter	0.9144	m	3	ft
Payload temperature	288.15	K	59	°F
Nozzle lift (weight)	1.95	kg	4.30	lb
Nozzle lift (force)	19.12	N	4.30	lbf
Burst altitude	26225.0681	m	86040.25	ft
Net lift (weight)	0.73	kg	1.62	lb
Net lift (force)	7.19	N	1.62	lbf
Time to burst altitude	64.82	min	64.82	min
Time to descend (use S.A. calculator)	33.30	min	33.30	min
Total mission time	98.12	min	98.12	min
Burst diameter	5.94	m	19.50	ft
Burst volume	109.94	m³	3882.42	ft³
Burst pressure	4522.85	pa	0.40	psi
Calculated descent time	68.85	min	68.85	min
Launch altitude	300	m		
Average ascent rate	6.4462887	m/s		
Calculated burst altitude	27400	m	89895.02	ft
Average Descent rate	13.7137137	m/s		

ASCENT	Second	Altitude above sea level		Atmospheric Pressure		Volume	Balloon diameter	Balloon lift	Net lift	Ascent velocity	Air Density	Helium
Time		m	ft	Pa	Psi	m³	m	kg	kg	m/s	kg/m³	kg/m³
10:35:00 AM	0	299.9232	984.0000315	97785.36051	14.10486104	1.85	1.524	1.948594606	0.132594606	1.570655855	1.23	
10:35:01 AM	1	301.4938559	989.153102	97767.15322	14.10180867	1.853678424	1.5240946	1.948957495	0.732957495	3.69258539	1.23	
10:35:02 AM	2	305.1864412	1001.267884	97724.36154	14.0946352	1.854490115	1.524317025	1.949810907	0.733810907	3.69419535	1.23	
10:35:03 AM	3	308.8806366	1013.387948	97681.56994	14.08746225	1.855302516	1.524539579	1.950665065	0.734665065	3.695805155	1.23	
10:35:04 AM	4	312.5764417	1025.513293	97638.77845	14.08028984	1.856115627	1.524762263	1.95151997	0.73551997	3.697414803	1.23	
10:35:05 AM	5	316.2738566	1037.64392	97595.98708	14.07311795	1.856929449	1.524985077	1.952375622	0.736375622	3.699024295	1.23	
10:35:06 AM	6	319.9728808	1049.779826	97553.19585	14.06594659	1.857743982	1.52520802	1.953232022	0.737232022	3.700633631	1.23	
10:35:07 AM	7	323.6735145	1061.921013	97510.40478	14.05877578	1.858559227	1.525431093	1.954089171	0.738089171	3.702242812	1.23	
10:35:08 AM	8	327.3757573	1074.06748	97467.61388	14.0516055	1.859375184	1.525654295	1.954947068	0.738947068	3.703851836	1.23	
10:35:09 AM	9	331.0796091	1086.219225	97424.82317	14.04443576	1.860191854	1.525877627	1.955805716	0.739805716	3.705460704	1.23	
10:35:10 AM	10	334.7850698	1098.376249	97382.03268	14.03726658	1.861009239	1.526101089	1.956665113	0.740665113	3.707063947	1.23	
10:35:11 AM	11	338.4921392	1110.53855	97339.24241	14.03009794	1.861827337	1.526324681	1.957525262	0.741525262	3.708677973	1.23	
10:35:12 AM	12	342.2008172	1122.706129	97296.45239	14.02292985	1.86264615	1.526548402	1.958386162	0.742386162	3.710286374	1.23	
10:35:13 AM	13	345.911036	1134.878985	97253.66264	14.01576232	1.863465679	1.526772253	1.959247815	0.743247815	3.711894618	1.23	
10:35:14 AM	14	349.6229982	1147.057117	97210.87317	14.00859535	1.864285924	1.526996234	1.96010922	0.74410922	3.713502707	1.23	
10:35:15 AM	15	353.3365009	1159.240526	97168.084	14.00142894	1.865106885	1.527220345	1.960973379	0.744973379	3.715110641	1.23	
10:35:16 AM	16	357.0516116	1171.429209	97125.29515	13.9942631	1.865928564	1.527444586	1.961837292	0.745837292	3.716718418	1.23	
10:35:17 AM	17	360.76833	1183.623168	97082.50664	13.98709782	1.86675096	1.527668957	1.96270196	0.74670196	3.718326039	1.23	
10:35:18 AM	18	364.486656	1195.822401	97039.71848	13.97993312	1.867574075	1.527893458	1.963567383	0.747567383	3.719933505	1.23	
10:35:19 AM	19	368.2065895	1208.026907	96996.93069	13.97276899	1.868397909	1.528118089	1.964433562	0.748433562	3.721540816	1.23	
10:35:20 AM	20	371.9281303	1220.236687	96954.1433	13.96560545	1.869222463	1.52834285	1.965300497	0.749300497	3.72314797	1.23	
10:35:21 AM	21	375.6512783	1232.45174	96911.35631	13.958444248	1.870047736	1.528567742	1.96616819	0.75016819	3.724754969	1.23	
10:35:22 AM	22	379.3760333	1244.672065	96868.56976	13.95128009	1.870873731	1.528792763	1.96703664	0.75103664	3.726361812	1.23	
10:35:23 AM	23	383.1023951	1256.897662	96825.78364	13.9441183	1.871700447	1.529017915	1.96790585	0.75190585	3.7279685	1.23	
10:35:24 AM	24	386.8303636	1269.12853	96782.99799	13.93695709	1.872527885	1.529243197	1.968775818	0.752775818	3.729575032	1.23	
10:35:25 AM	25	390.5593986	1281.364669	96740.21282	13.92979548	1.873356045	1.529468609	1.969646546	0.753646546	3.731181408	1.23	
10:35:26 AM	26	394.29112	1293.606078	96697.42815	13.92263647	1.874184929	1.529694152	1.970518034	0.754518034	3.732787629	1.23	
10:35:27 AM	27	398.0239077	1305.852757	96654.64399	13.91547706	1.875014536	1.529919825	1.971390284	0.755390284	3.734393694	1.23	
10:35:28 AM	28	401.7583014	1318.104705	96611.86037	13.90831825	1.875844868	1.530145628	1.972263295	0.756263295	3.735999604	1.23	
10:35:29 AM	29	405.494301	1330.361922	96569.0773	13.90116004	1.876675925	1.530371562	1.973137068	0.757137068	3.737605359	1.23	
10:35:30 AM	30	409.2319063	1342.624408	96526.2948	13.89400245	1.877507708	1.530597626	1.974011604	0.758011604	3.739210958	1.23	
10:35:31 AM	31	412.9711173	1354.89216	96483.51288	13.88684547	1.878340217	1.530823821	1.974886904	0.758886904	3.740816401	1.23	
10:35:32 AM	32	416.7119337	1367.16518	96440.73157	13.87968911	1.879173452	1.531050146	1.975762968	0.759762968	3.74242169	1.23	
10:35:33 AM	33	420.4543554	1379.443467	96397.95089	13.87253336	1.880007415	1.531276602	1.976639796	0.760639796	3.744026823	1.23	
10:35:34 AM	34	424.1983822	1391.727202	96355.17084	13.86537824	1.880842106	1.531503189	1.977517391	0.761517391	3.7456318	1.23	
10:35:35 AM	35	427.944014	1404.015839	96312.39146	13.85822374	1.881677526	1.531729906	1.978395751	0.762395751	3.747236623	1.23	

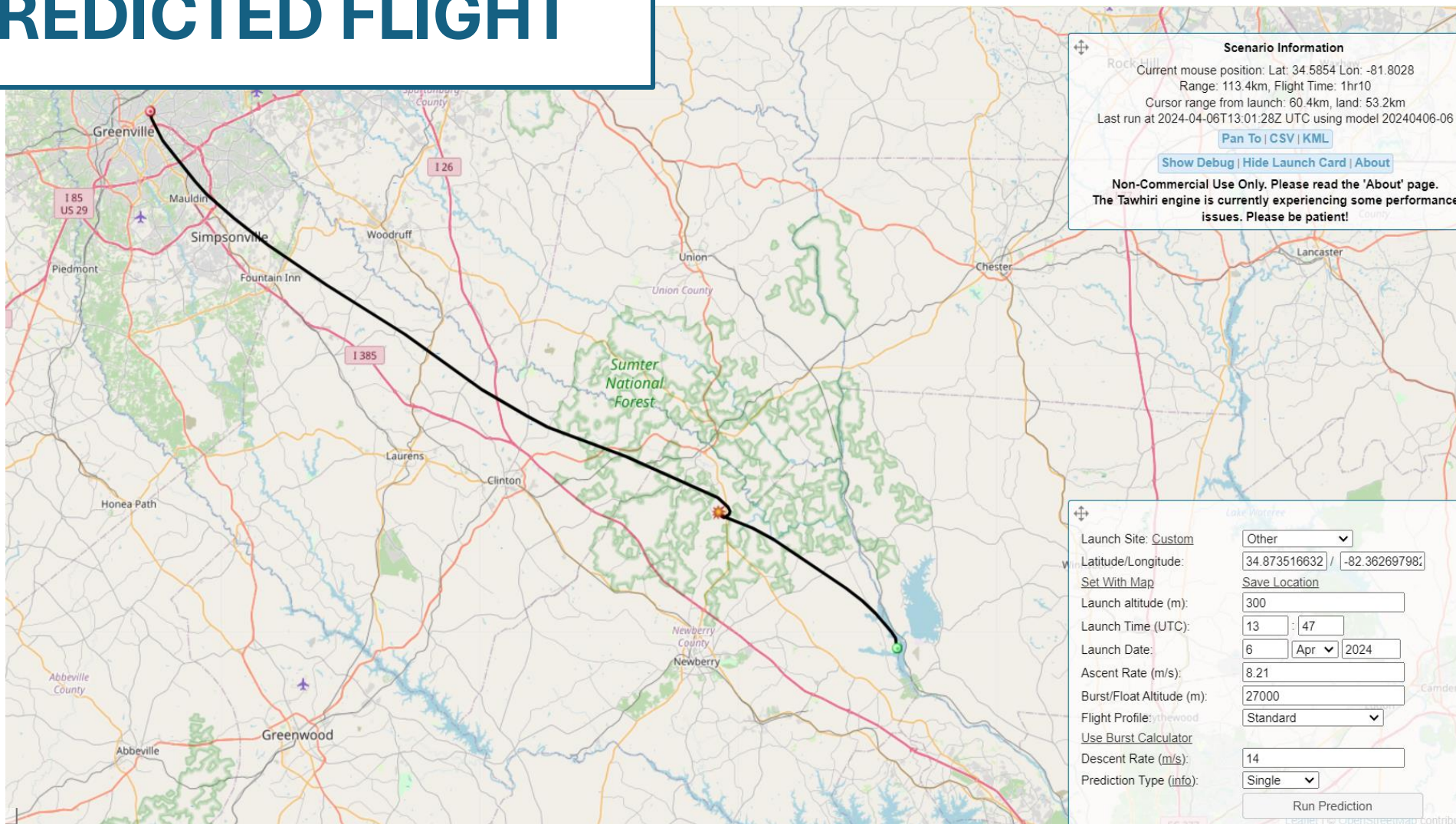
Dynamics Predictions

Metric	Predicted	Actual
Flight time	1 h 38 min	1h 19 min
Ascent rate	6.45m/s	6.34m/s
Burst altitude	27,400 m	18,700 m
Descent rate	13.7 m/s	9.24 m/s

Imperial	Predicted	Actual
Flight time	1 h 38 min	1h 19 min
Ascent rate	21.2 ft/s	20.8 ft/s
Burst altitude	89,900 ft	61,400 ft
Descent rate	45.0 ft/s	30.3 ft/s

Tracking

PREDICTED FLIGHT



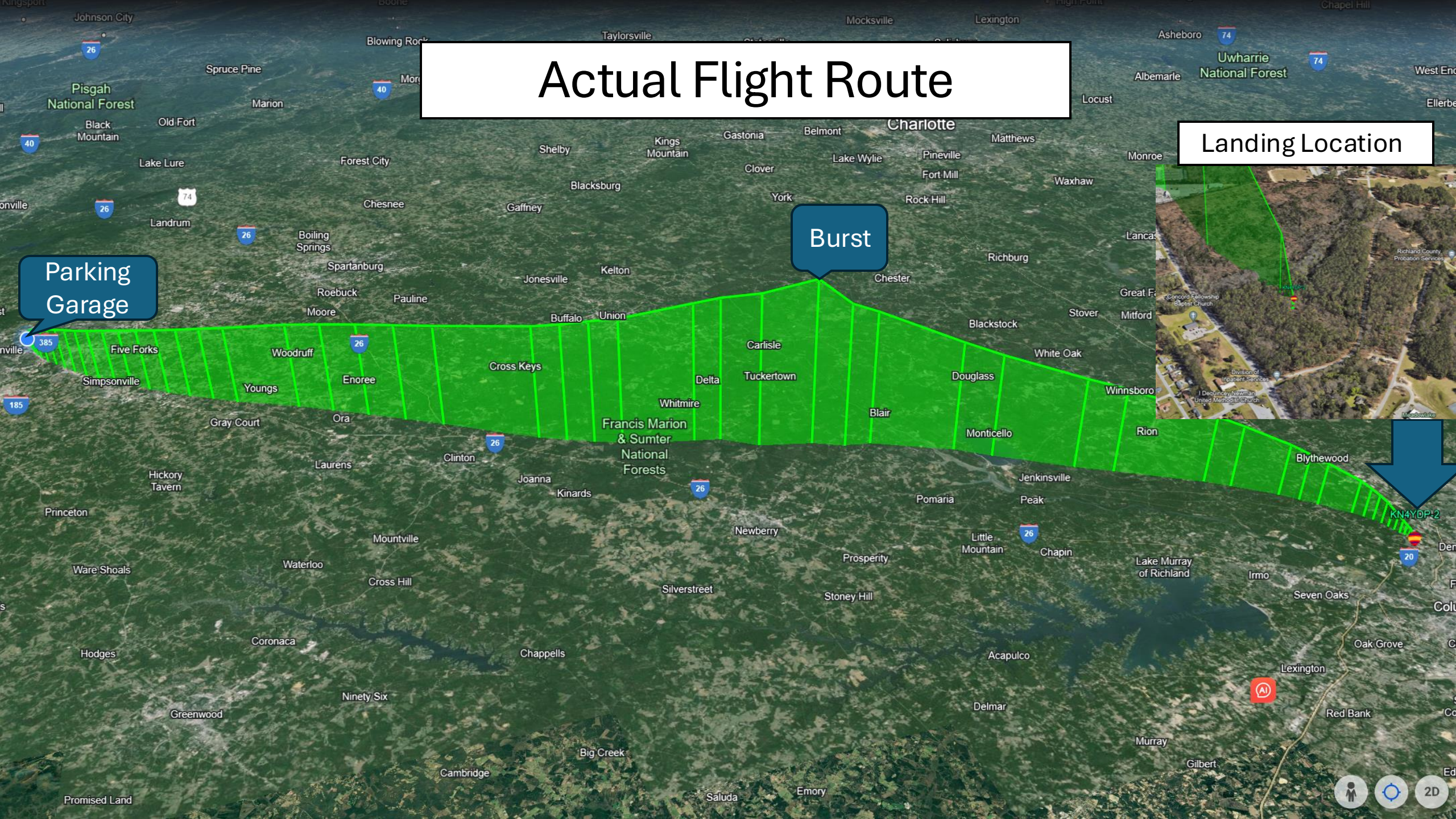
Prediction Source: CUSF SondeHub

Actual Flight Route

Landing Location

Parking
Garage

Burst



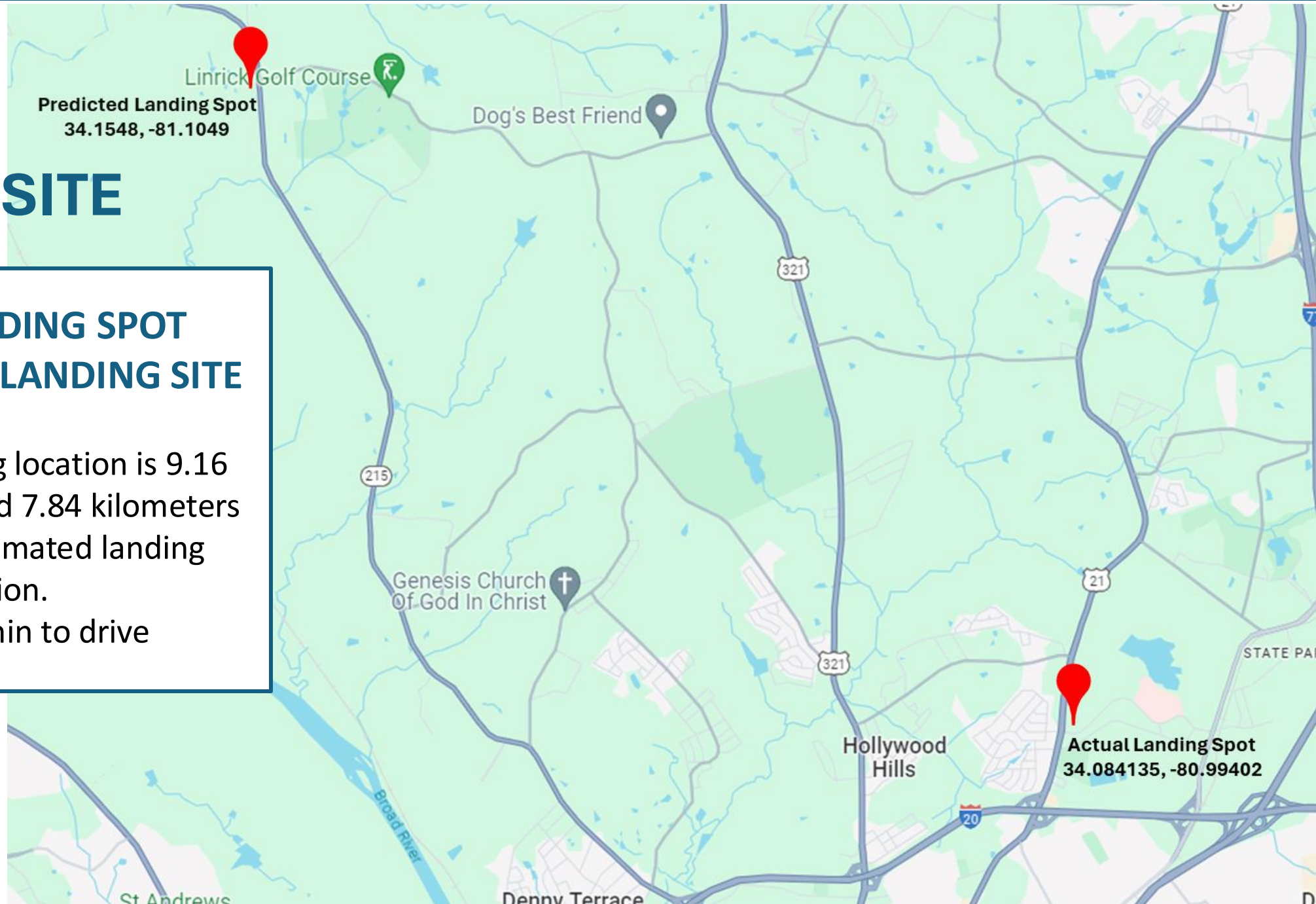
2D

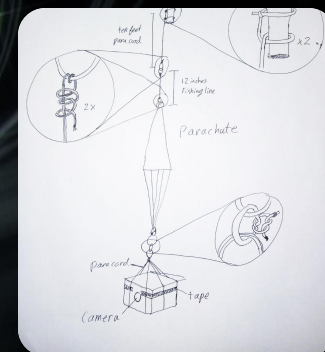
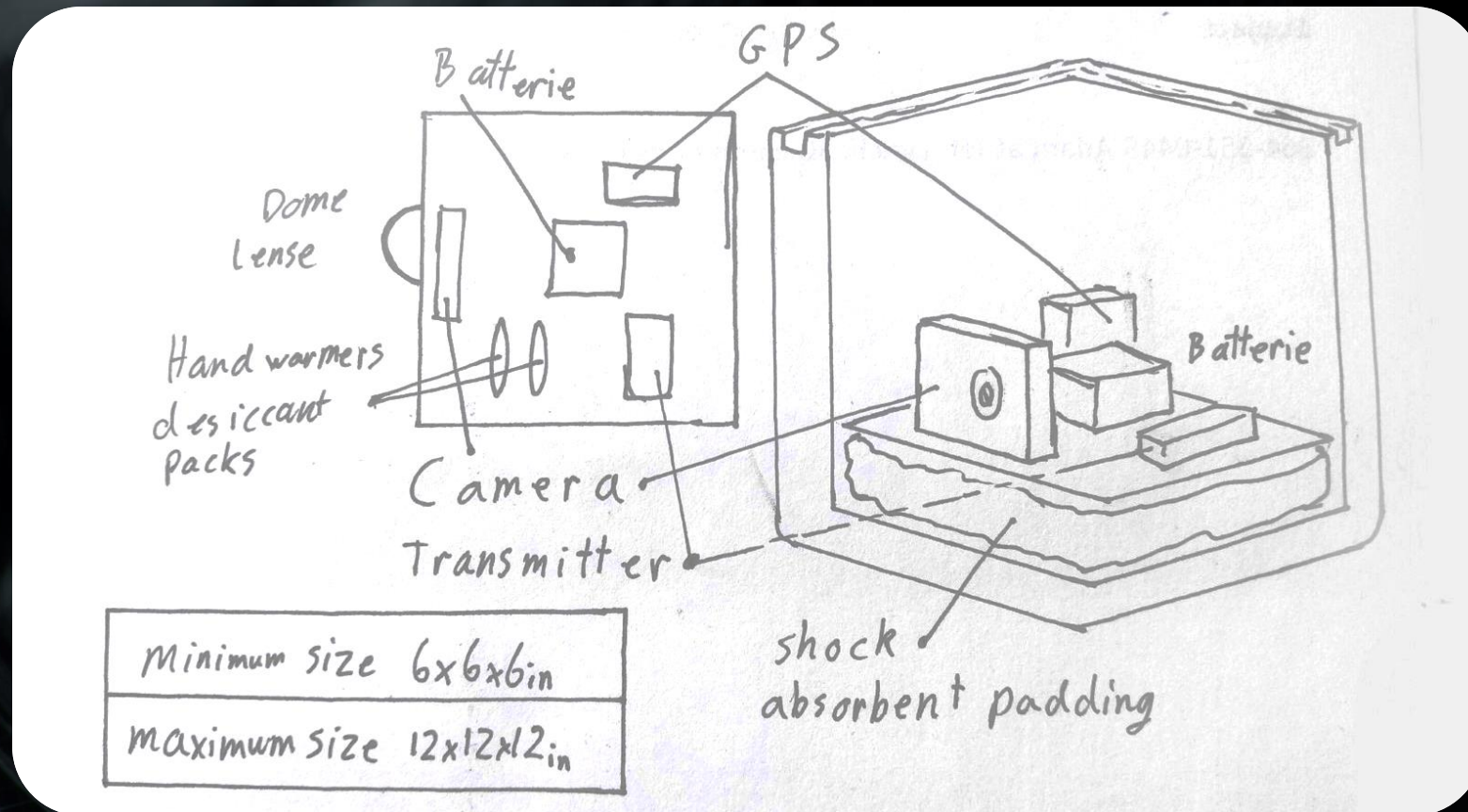
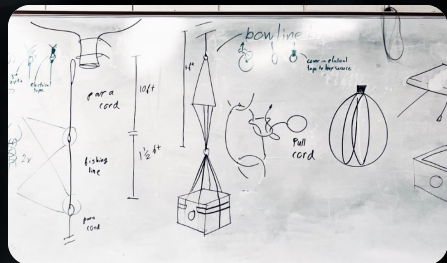
LANDING SITE

ACTUAL LANDING SPOT VS. PREDICTED LANDING SITE

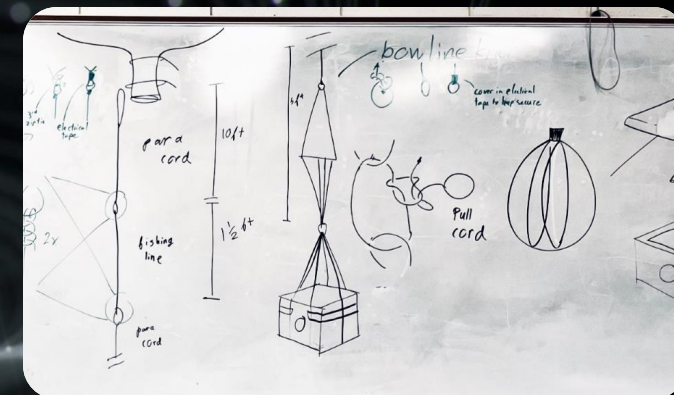
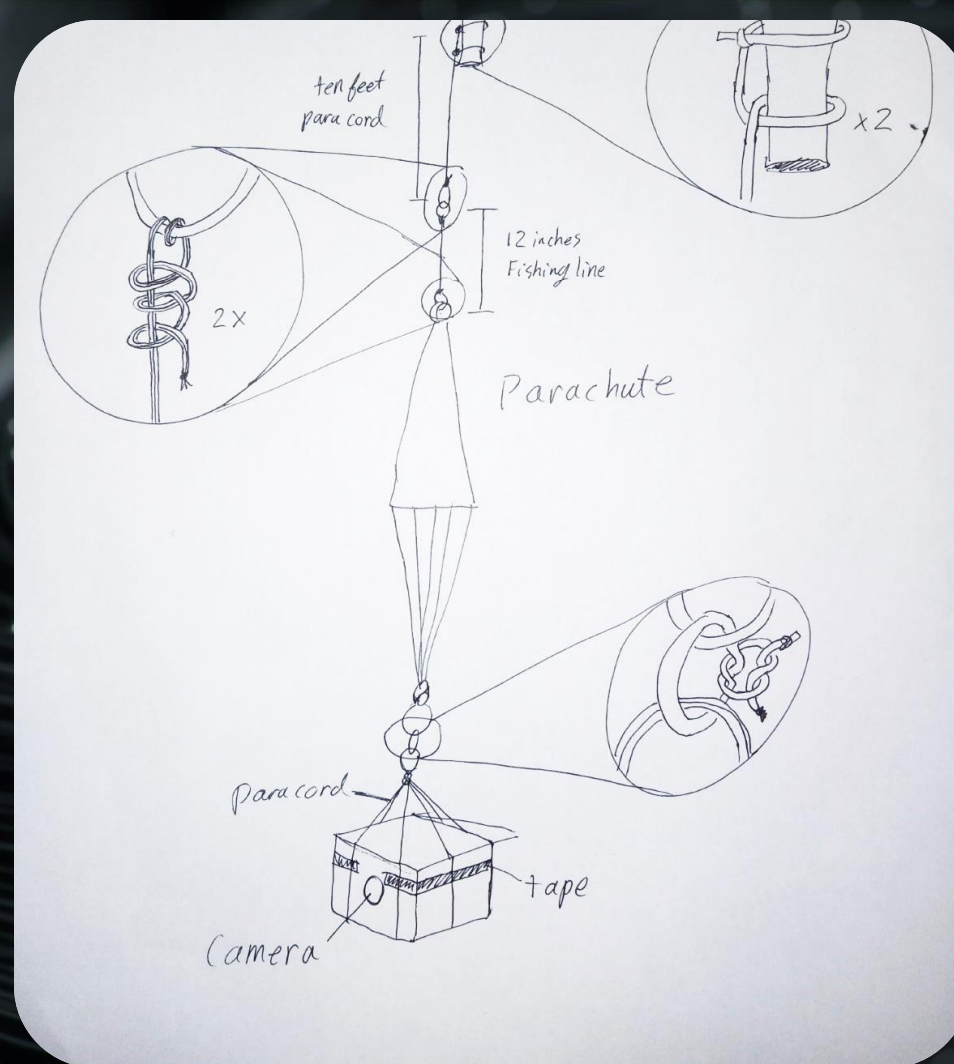
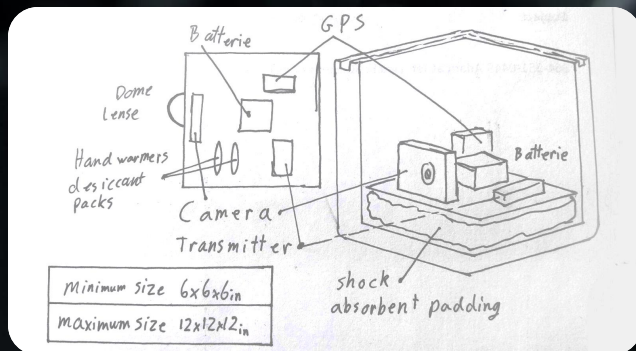
The actual landing location is 9.16 kilometers east and 7.84 kilometers south of the estimated landing location.

It takes 15 min to drive

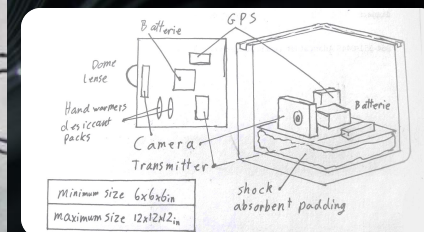
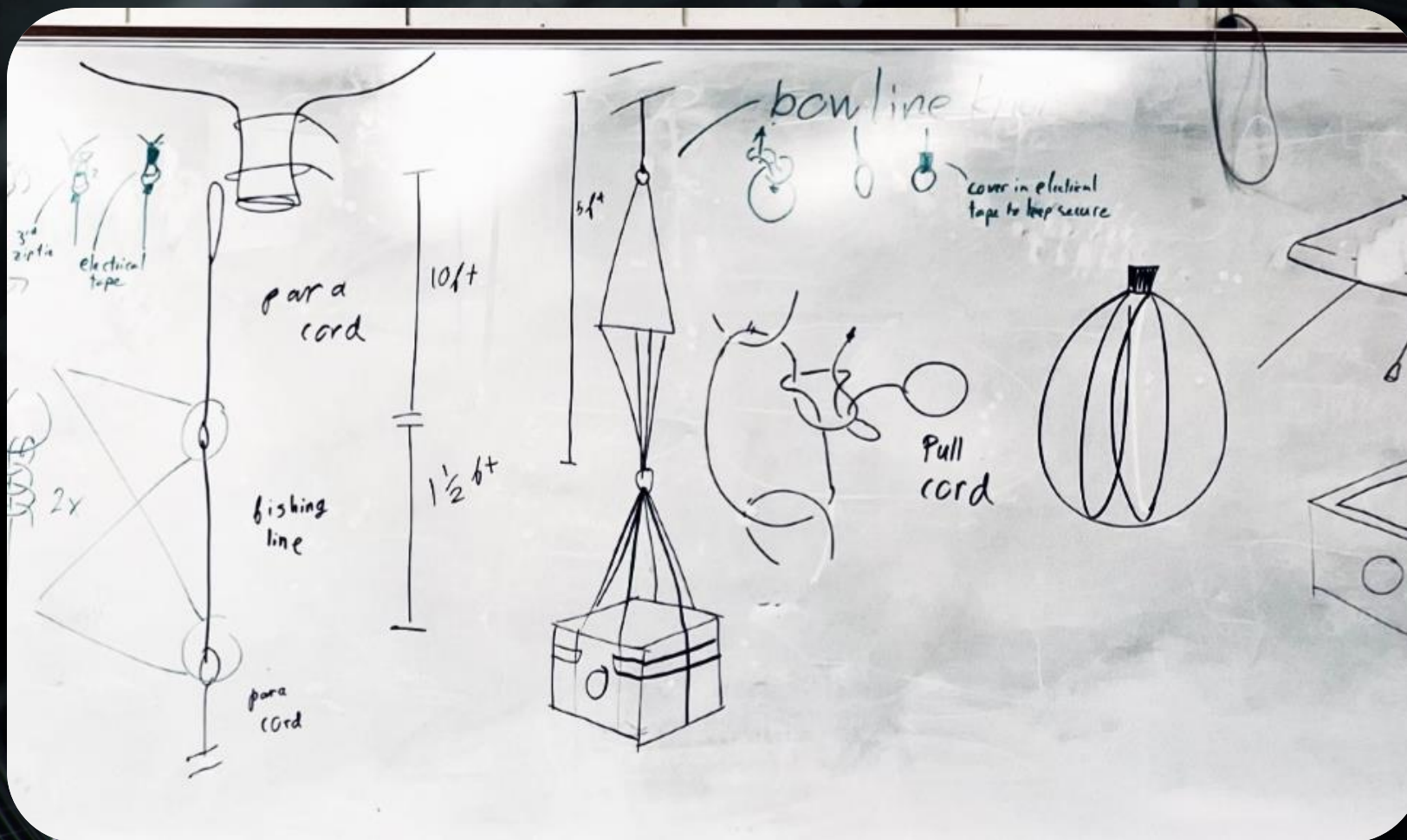
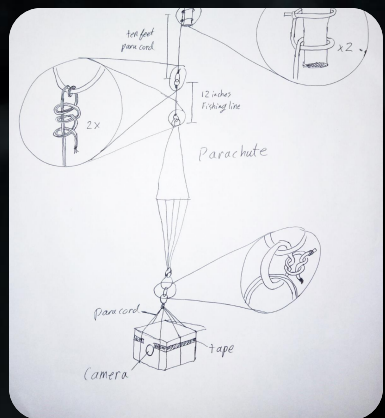




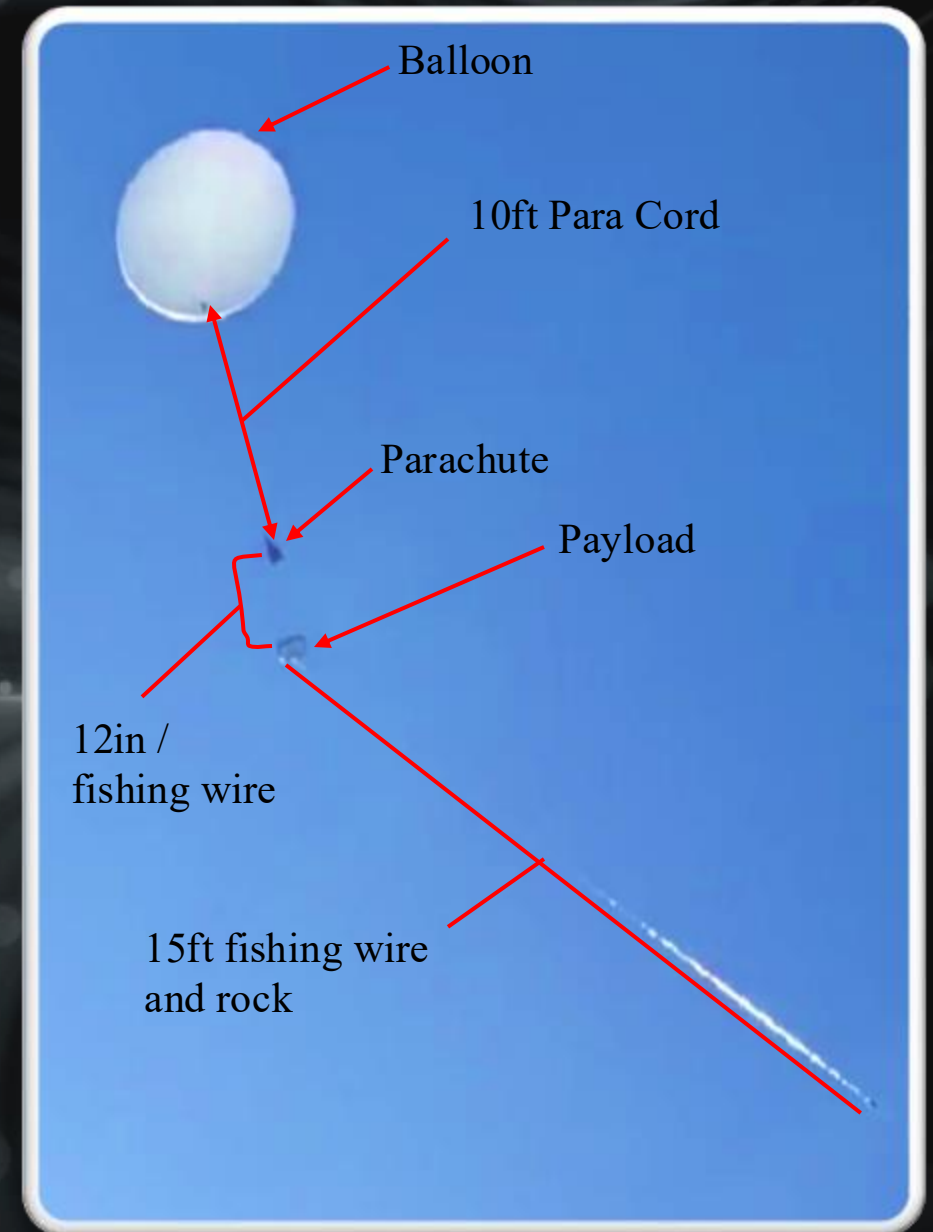
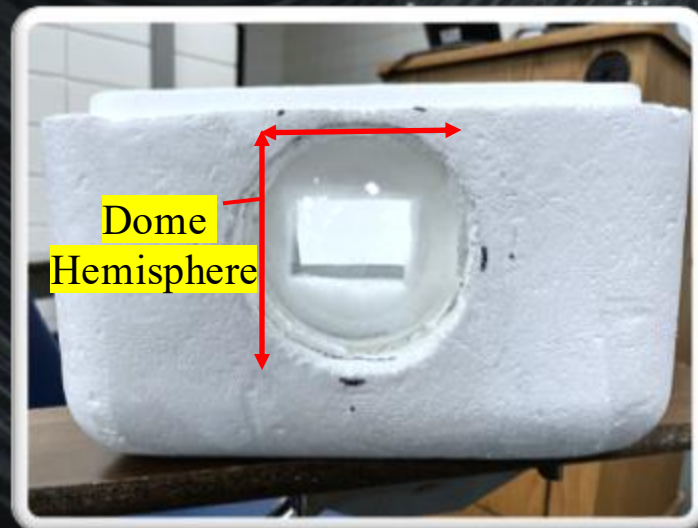
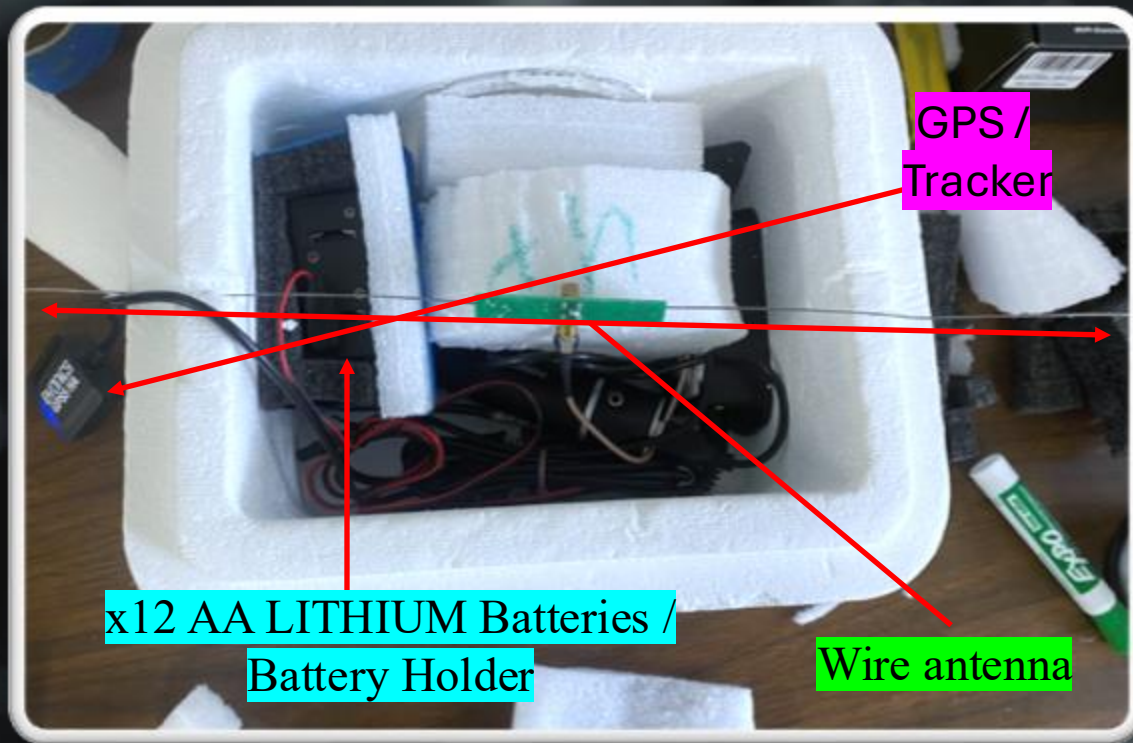
Survivability

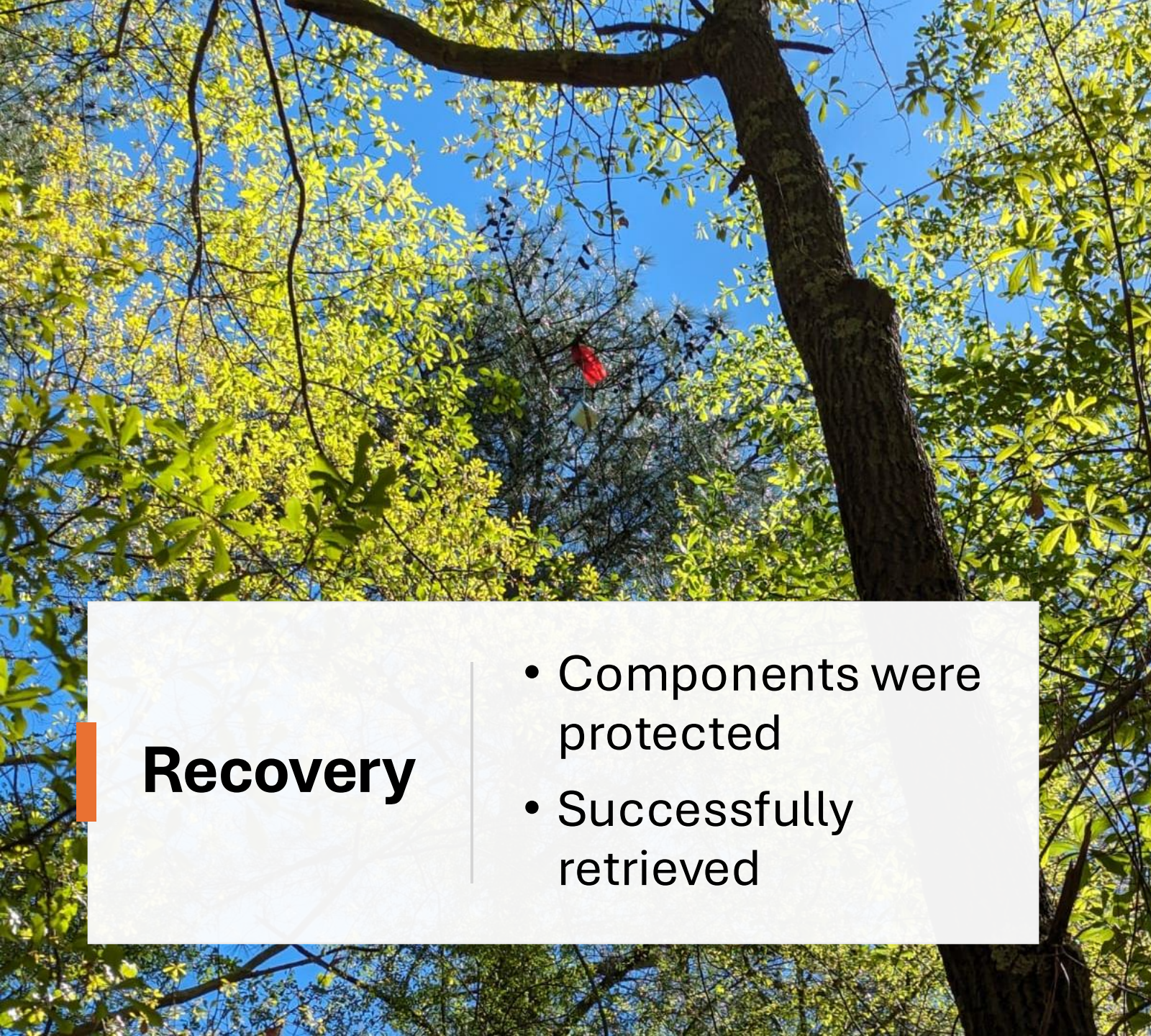


Survivability



Survivability



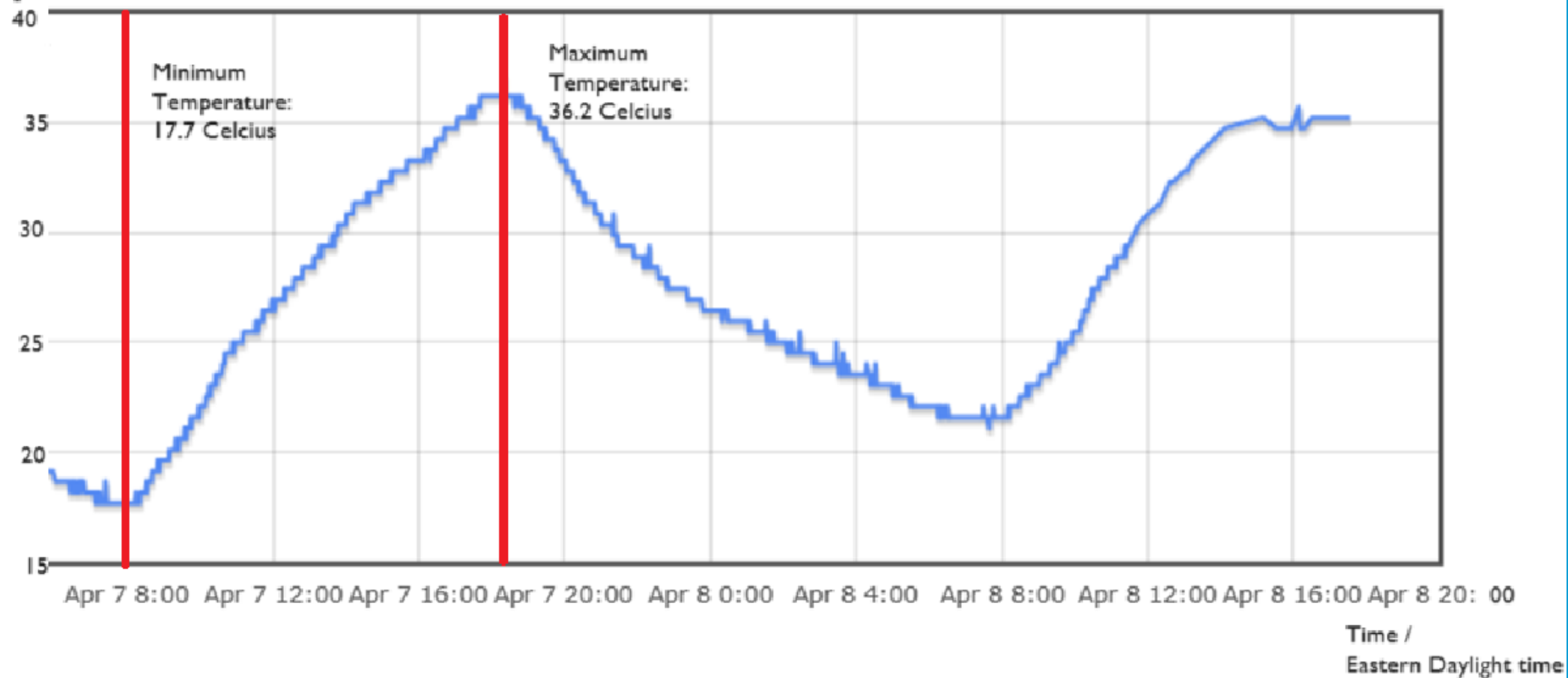


Recovery

- Components were protected
- Successfully retrieved



Temperature
[Celcius]



Electronics

Issues fixed before launch day

-Power source stability

-Bad transmission due to tangled cords

Overall tracking was a huge success.

-We were able to continue tracking after two days.

Video was corrupted.

GPS Receiver

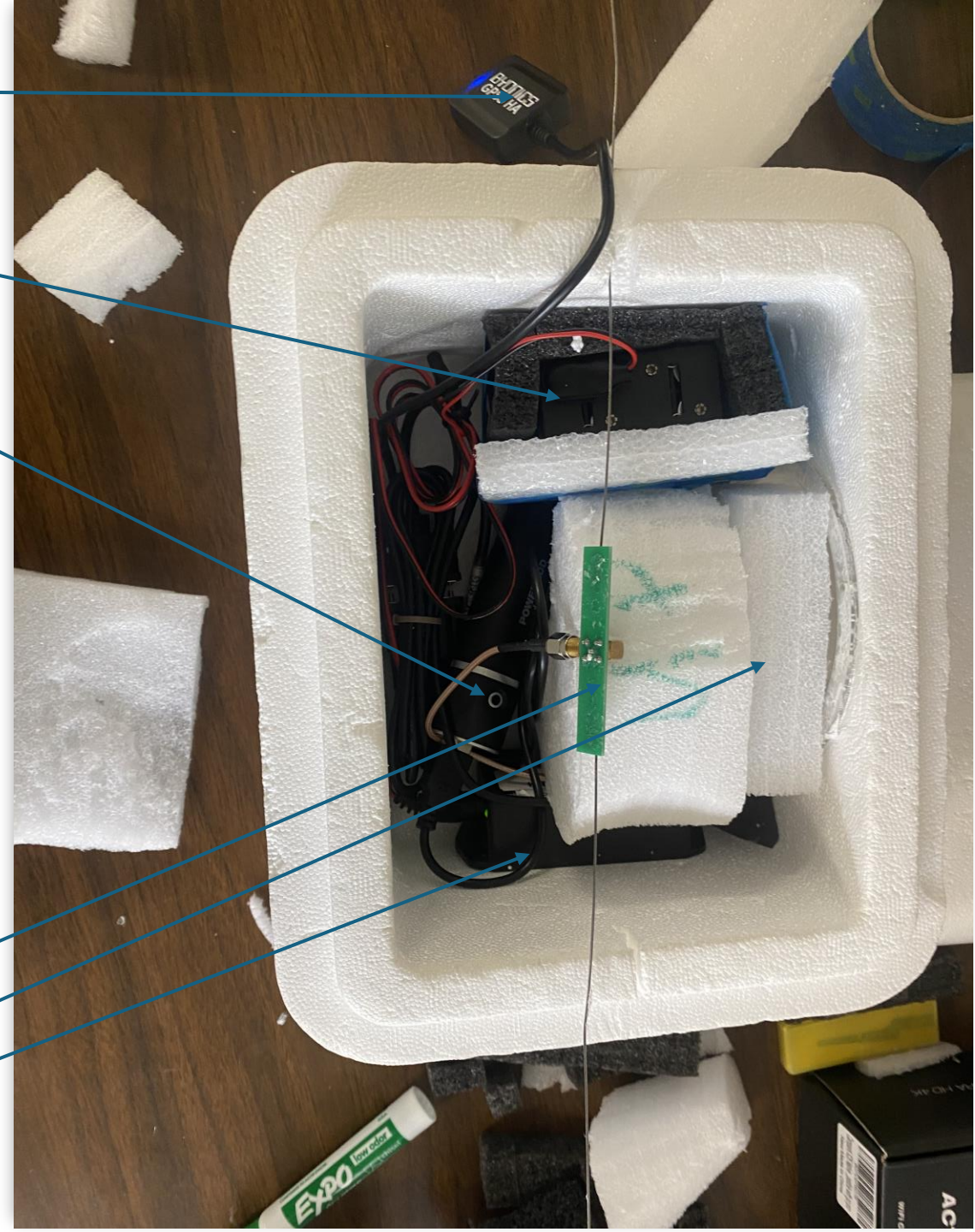
Batteries

Power
Bank

Antenna

Camera

APRS
Transmitter



Lessons Learned

- Communication
- Testing Every Component



Thank You